



OPEN Potential utility of salivary gland amyloid uptake as a surrogate marker for adipose tissue and amyloid-beta dynamics based on its association with BMI and cerebral amyloid burden

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Recent studies have noted an unexpected inverse correlation between body mass index (BMI) and cerebral amyloid-beta (A β) deposition, suggesting a potential involvement of adipose tissue in AD pathology. Given the adipose-rich nature of salivary glands, we investigated whether A β uptake in the salivary glands is associated with the link between BMI and cerebral A β deposition using ¹⁸F-Florbetaben PET/CT. We analyzed data from 187 participants, classified into A β ⁻ cognitively normal (A β ⁻CN), A β ⁺ mild cognitive impairment (A β ⁺MCI), and A β ⁺ AD dementia (A β ⁺D) groups. BMI, standardized uptake value ratio of salivary glands (SUVR_{SG}), and brain (SUVR_B) were quantified, and their associations were examined using Pearson correlation and multiple regression analyses. Higher BMI was correlated with increased SUVR_{SG} ($r = 0.231$, $p = 0.0019$) and decreased SUVR_B ($r = -0.247$, $p = 0.0009$). SUVR_{SG} and SUVR_B were negatively correlated ($r = -0.188$, $p = 0.0102$). Multiple regression analysis identified BMI as an independent predictor of composite brain SUVR ($\beta = -0.0168$, $p = 0.0046$), while SUVR_{SG} was marginally significant ($\beta = -0.0714$, $p = 0.0701$). Regionally, both BMI and SUVR_{SG} were significantly associated with lower SUVR_B in the striatum, precuneus, and cingulate cortex. Consistent with previous studies, BMI exhibited an inverse correlation with SUVR_B. Notably, SUVR_{SG} was negatively associated with SUVR_B while demonstrating a positive correlation with BMI. Interpreting SUVR_{SG} as a fat-related marker, these findings suggest that A β sequestration in adipose tissue may contribute to lower cerebral A β accumulation. Additional research is necessary to elucidate the underlying mechanism.

Keywords Alzheimer's disease, Amyloid PET, Standardized uptake value ratio of salivary glands (SUVR_{SG}), Body mass index, Adipose tissue

Alzheimer's disease (AD) is a progressive neurodegenerative disorder marked by cognitive deterioration and the pathological accumulation of amyloid-beta (A β) plaques in the brain¹. The aggregation of A β is a key pathological hallmark of AD, contributing to synaptic dysfunction, neuroinflammation, and neuronal degeneration, thus making it a crucial focus in understanding AD pathogenesis².

The relationship between body mass index (BMI) and AD has been extensively investigated, revealing intriguing temporal variations across the lifespan. While midlife obesity correlates with elevated AD risk^{3,4} and has been associated with enhanced A β deposition and neurodegeneration, the late-life BMI profile presents a contrasting pattern. Specifically, reduced BMI in older adults demonstrates a significant correlation with increased cerebral A β burden, particularly within the 1–2 years preceding positron emission tomography assessment⁵. This pattern is further complicated by the fact that increased visceral adiposity in the elderly

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corresponds with cortical thinning—a negative marker of AD progression—most prominently in the temporal, parietal, and cingulate regions⁶.

These findings collectively suggest that adipose tissue and related metabolic factors may substantially influence A β accumulation dynamics and neurodegenerative processes, though the precise mechanistic pathways underlying these relationships remain undefined. Studies on the association of adipose tissue metabolism with AD pathology indicate that glucose metabolism measured from visceral adipose tissue was significantly associated with cerebral A β burden, suggesting that metabolic dysfunction of adipose tissue could contribute to AD development⁷. Similarly, confirming A β accumulation within adipose tissue is expected to be crucial in elucidating mechanistic pathways, although not yet directly established.

Salivary glands (SGs) are a distinct type of adipose-enriched tissue that may provide valuable insights into the undefined adipose tissue-A β accumulation mechanism. SGs have demonstrated a potential metabolic link with BMI⁸ and MRI-based fat quantification studies reveal that fat content in the parotid and submandibular glands increases in proportion to BMI⁹. Furthermore, obesity has been linked to decreased salivary flow rates, emphasizing the broader metabolic implications of SG function¹⁰. Given their adipose-rich nature and metabolic activity, SGs may serve as a surrogate indicator of adipose tissue involvement in systemic amyloid dynamics.

The degree of brain amyloid accumulation can be assessed via molecular imaging techniques using positron emission tomography (PET) with various amyloid-specific radiotracers, including ¹¹C-Pittsburgh compound B (PiB)¹¹, ¹⁸F-florbetapir¹², ¹⁸F-florbetaben¹³, and ¹⁸F-flutemetamol¹⁴. Notably, salivary glands, particularly the parotid glands, are within the field of view in amyloid PET scans, allowing for their assessment without additional imaging protocols or modifications. This accessibility makes them a convenient target for evaluating amyloid uptake in adipose tissue.

This study not only quantifies brain uptake but also evaluates amyloid uptake in the salivary glands using amyloid PET imaging. Our primary objective is to determine whether BMI is associated with amyloid retention in SGs and whether this uptake is linked to brain A β burden. Through this examination, we aim to verify the potential of salivary gland amyloid uptake as a surrogate for adipose tissue-A β accumulation mechanism, thereby enhancing our understanding of how adipose tissue influences brain amyloid deposition and AD pathology.

Materials and methods

Subjects

This study included participants who underwent ¹⁸F-Florbetaben PET/CT imaging at Ewha Womans University Mokdong Hospital between November 2018 and February 2023. Three diagnostic groups encompassing cognitively unimpaired individuals without amyloid deposition (A β -CN; NC group), individuals with mild cognitive impairment and amyloid deposition (A β +MCI; MCI group), and individuals with AD dementia (A β +D; AD group) were included in the analysis. The diagnostic criteria for MCI and AD were based on those proposed by the National Institute on Aging and Alzheimer's Association (NIA-AA)¹⁵. A β status was determined based upon ¹⁸F-Florbetaben PET scan results¹⁶.

Demographic and clinical data were collected for all participants, including sex, age, years of education, weight, height, BMI, and APOE4 carrier status. BMI was calculated as weight (kg) divided by height squared (m²). Cognitive function was assessed using the Mini-Mental State Examination (MMSE) and the Clinical Dementia Rating (CDR) scale.

This study was conducted in accordance with the Declaration of Helsinki (2013 version) and approved by the Institutional Review Board (IRB) of Ewha Womans University Mokdong Hospital (EUMC IRB 2025-01-003), and the requirement for informed consent was waived by the IRB of Ewha Womans University Mokdong Hospital.

¹⁸F-Florbetaben PET/CT image acquisition

The radiotracer ¹⁸F-FBB (Neuraceq™) was produced by DuChemBio Co., Ltd. (Seoul, Korea) in compliance with the Korean Ministry of Food and Drug Safety (MFDS). After being delivered to our institutional PET center, all ¹⁸F-FBB PET/CT images were processed following our institution's standardized protocol. PET images were acquired in three-dimensional (3D) list-mode using a specialized PET/CT scanner (Biograph mCT, Siemens) with 20-minute static scans obtained from 90 to 110 min after a bolus injection of 306.16 $\pm$ 17.85 MBq ¹⁸F-FBB. Additionally, a spiral CT scan of the brain was conducted with parameters of 120 kV, 30mAs, and a slice thickness of 1.0 mm. The resulting CT data were used for attenuation correction to the PET emission scans. The participant's head was immobilized by head-holder and vacuum-based fixation equipment to reduce motion artifacts. The PET images were reconstructed into a 400 $\times$ 400 $\times$ 109 matrix (voxel 1.02 $\times$ 1.02 $\times$ 2.03mm³) using the integrated 3D ordered subset expectation maximization algorithm, configured with 4 iterations, 12 subsets, and a 5-mm Gaussian smoothing filter. The reconstructed data were then combined into single-frame images.

Analysis of ¹⁸F-Florbetaben PET/CT images

Brain standardized uptake value ratio (SUVR_B) values were extracted using Neurophet SCALE PET 2.0 software (Neurophet Inc., South Korea), an artificial intelligence (AI)-based software, provides automated quantification by directly processing reconstructed DICOM (Digital Imaging and Communications in Medicine) PET images co-registered on T1w images^{17,18}. Automatic quantitative analyses were performed using volumes of interest (VOIs) defined in each individual's native T1w space, based on the Desikan-Killiany atlas¹⁹. For the SUVR quantification, we selected target regions, including cortical (frontal, precuneus, cingulate, lateral parietal, lateral temporal, and occipital cortices) and subcortical (striatum) regions, as well as a composite region encompassing all these target regions. The whole cerebellum was used as the reference region.

Salivary glands standardized uptake value ratio (SUVR_{SG}) values were extracted according to the methodology described in a previous study²⁰ with minor modifications. For each parotid gland, a sphere volume of interest

(VOI) was manually delineated by the investigator, and the final VOI was determined using a 40% threshold of the maximum pixel value, which was automatically applied by MIM software (MIM Software Inc., Cleveland, OH). The mean SUV values from both glands were averaged for analysis. The relative uptake ratio (SUV_{SG}) was defined as the mean SUV of the salivary glands divided by the SUV of the cerebellum. The volume of each parotid gland (Vol_{SG}) was estimated using the target volume within the VOI, applying the 40% threshold.

Neuropsychological assessment

The Mini-Mental State Examination (MMSE) was performed for neuropsychological assessment. MMSE includes a set of questions that evaluate five areas of cognitive function: orientation, registration, attention and calculation, recall and language. The test has a maximum score of 30 points. A score of 24 or higher suggests normal cognitive function, while scores ranging from 19 to 23 indicate mild cognitive impairment. Scores between 10 and 18 are classified as moderate cognitive impairment, and scores below 9 reflect severe cognitive impairment.

The Clinical Dementia Rating (CDR) was also administered to assess the severity of Alzheimer’s disease. The CDR consists of semistructured interviews conducted with both the patient and an informant. During the interview, the informant rates the patient’s cognitive performance across six domains: Memory, Orientation, Judgment and Problem Solving, Community Affairs, Home and Hobbies and Personal Care. The CDR is scored on a five-point scale: 0 (none), 0.5(questionable), 1(mild), 2(moderate) and 3(severe), with higher scores indicating greater severity of cognitive impairment. CDR 0 means no dementia, while higher scores correspond to varying degrees of dementia, ranging from questionable to severe.

Statistical analysis

All statistical analyses were performed using Rex software (version 3.7.0.1, RexSoft, Seoul, Korea), with statistical significance set at $p < 0.05$ (two-tailed). Group differences in demographic characteristics and PET imaging features were assessed using one-way ANOVA for normally distributed variables and the Kruskal-Wallis test for non-normally distributed variables. Categorical variables were compared using chi-square tests. The relationships between BMI, SUV_{RB} , SUV_{SG} , and salivary gland volume (Vol_{SG}) were analyzed using Pearson correlation coefficient. Multiple regression analysis was performed to evaluate the independent effects of BMI and SUV_{SG} with age as a covariate using the enter method. A Bonferroni-corrected threshold of $p < 0.0071$ ($0.05/7$) was applied for the regional SUV analyses across seven target regions.

Results

Subjects

This study included 187 participants who underwent ^{18}F -Florbetaben PET/CT imaging. Participants were categorized into three diagnostic groups: amyloid-negative cognitively unimpaired individuals ($A\beta^{-}CN$, $n = 114$), amyloid-positive individuals with mild cognitive impairment ($A\beta^{+}MCI$, $n = 45$), and amyloid-positive individuals with dementia ($A\beta^{+}D$, $n = 28$). Demographic and clinical characteristics of each group are summarized in Table 1. The median age of participants was 72 years (interquartile range: 68–77), with no statistically significant difference among the groups ($p = 0.1559$). BMI was significantly different among the groups ($p = 0.0292$), with the $A\beta^{-}CN$ group having the highest median BMI (24.21, interquartile range: 22.43–27.08), whereas the $A\beta^{+}MCI$ (22.9, interquartile range: 21.33–24.84) and $A\beta^{+}D$ (23.27, interquartile range: 21.43–24.77) groups exhibited lower values. The prevalence of APOE4 carriers was notably higher in the $A\beta^{+}MCI$ (66.67%) and $A\beta^{+}D$ (71.43%) groups compared to the $A\beta^{-}CN$ group (11.4%) ($p < 0.001$). Cognitive function, assessed via MMSE and CDR scores, demonstrated significant group differences ($p < 0.001$), with the $A\beta^{+}D$ group exhibiting the lowest MMSE score (18.5, interquartile range: 15–20) and the highest CDR score (1.0, interquartile range: 1–1).

	Clinical diagnosis				
	Total	$A\beta^{-}CN$	$A\beta^{+}MCI$	$A\beta^{+}D$	<i>p</i> -value
Demographic characteristics					
Subject (n)	187	114	45	28	
Age (years)	72 (68, 77)	72 (67, 76)	73 (70, 79)	73.5 (69.75, 78.25)	0.1559
Sex (female: male)	128:59	83:31	22:23	23:5	0.0033*
BMI	23.71 (22, 26.26)	24.21 (22.43, 27.08)	22.9 (21.33, 24.84)	23.27 (21.43, 24.77)	0.0292*
Clinical characteristics					
APOE4 carrier (%)	33.69	11.4	66.67	71.43	$p < 0.001^{*}$
MMSE score	28 (25, 29)	29 (28, 30)	25 (22, 27.25)	18.5 (15, 20)	$p < 0.001^{*}$
CDR score	0.5 (0, 0.5)	0.5 (0, 0.5)	0.5 (0.5, 0.5)	1 (1, 1)	$p < 0.001^{*}$

Table 1. Demographic data and clinical characteristics. Continuous variables are presented as mean ± standard deviation for normally distributed data and as median (interquartile range) for non-normally distributed data. Categorical variables are expressed as counts and percentages. BMI body mass index, MMSE mini-mental state examination, CDR clinical dementia rating, $A\beta^{-}CN$ amyloid-negative cognitively unimpaired, $A\beta^{+}MCI$ amyloid-positive mild cognitive impairment, $A\beta^{+}D$ amyloid-positive dementia.

	Clinical diagnosis				
	Total	A β ⁻ CN	A β ⁺ MCI	A β ⁺ D	p-value
PET indices					
SUVR _{SG}	2.16 (1.91, 2.47)	2.23 (1.97, 2.58)	2.04 (1.88, 2.25)	2.09 (1.78, 2.3)	0.0049*
Vol _{SG}	23.91 (17.11, 31.13)	23.27 (16.86, 30.19)	24.86 (17.45, 32.23)	23.26 (14.83, 30.28)	0.6023
SUVR _B	1.37 ± 0.25	1.19 ± 0.05	1.62 ± 0.19	1.67 ± 0.17	p < 0.001*
SUVR _F	1.36 ± 0.27	1.18 ± 0.06	1.63 ± 0.21	1.67 ± 0.19	p < 0.001*
SUVR _{PCu}	1.3 (1.24, 1.77)	1.26 (1.21, 1.29)	1.86 (1.61, 2.01)	1.9 (1.7, 2.06)	p < 0.001*
SUVR _{Cg}	1.46 ± 0.29	1.26 ± 0.07	1.76 ± 0.22	1.79 ± 0.23	p < 0.001*
SUVR _{LP}	1.38 ± 0.27	1.20 ± 0.07	1.65 ± 0.19	1.72 ± 0.18	p < 0.001*
SUVR _{LT}	1.27 ± 0.23	1.11 ± 0.05	1.49 ± 0.19	1.56 ± 0.16	p < 0.001*
SUVR _{OC}	1.28 (1.21, 1.42)	1.23 (1.19, 1.28)	1.43 (1.32, 1.56)	1.5 (1.37, 1.62)	p < 0.001*
SUVR _{Str}	1.47 ± 0.2	1.36 ± 0.09	1.66 ± 0.18	1.64 ± 0.18	p < 0.001*

Table 2. PET indices by clinical diagnosis. Continuous variables are presented as mean ± standard deviation for normally distributed data and as median (interquartile range) for non-normally distributed data. Categorical variables are expressed as counts and percentages. SUVR_{SG} standardized uptake value ratio of salivary gland, Vol_{SG} volume of salivary gland, SUVR_B global standardized uptake value ratio of brain, SUVR_F standardized uptake value ratio of frontal cortex, SUVR_{PCu} standardized uptake value ratio of precuneus, SUVR_{Cg} standardized uptake value ratio of cingulate cortex, SUVR_{LP} standardized uptake value ratio of lateral parietal cortex, SUVR_{LT} standardized uptake value ratio of lateral temporal cortex, SUVR_{OC} standardized uptake value ratio of occipital cortex, SUVR_{Str} standardized uptake value ratio of striatum, A β ⁻CN amyloid-negative cognitively unimpaired, A β ⁺MCI amyloid-positive mild cognitive impairment, A β ⁺D amyloid-positive dementia.

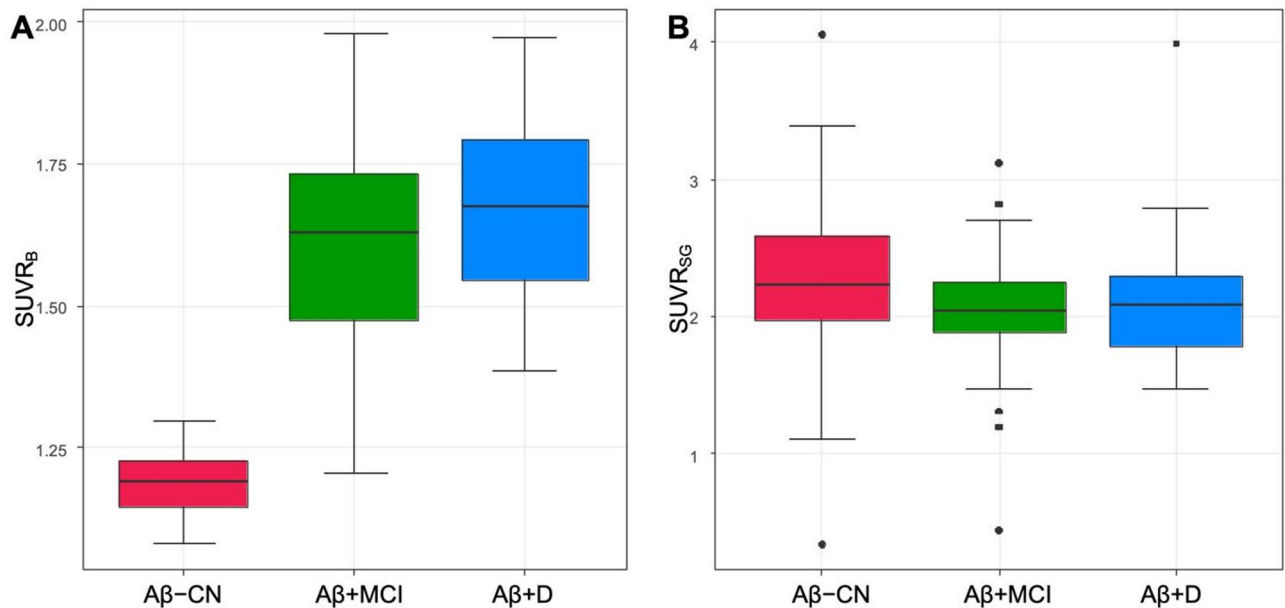


Fig. 1. Group-wise comparison between brain and salivary gland amyloid uptake. (A) The Standardized Uptake Value Ratio of the Brain (SUVR_B) significantly differed among groups ($p < 0.001$), with the highest values observed in the A β ⁺MCI and A β ⁺D groups, indicating increased cerebral amyloid deposition in individuals with cognitive impairment. (B) The Standardized Uptake Value Ratio of the Salivary Glands (SUVR_{SG}) also showed significant group differences ($p = 0.0049$), with the highest uptake in the A β ⁻CN group. Data are presented as mean ± standard deviation, and statistical significance was determined using one-way ANOVA.

PET indices across diagnostic groups are presented in Table 2. SUVR_B significantly increased across groups ($p < 0.001$, Fig. 1A), with the highest value observed in the A β ⁺D group (1.67 ± 0.17). Regional SUVR values, including those for the frontal cortex (SUVR_F), precuneus (SUVR_{PCu}), cingulate cortex (SUVR_{Cg}), lateral parietal cortex (SUVR_{LP}), lateral temporal cortex (SUVR_{LT}), occipital cortex (SUVR_{OC}), and striatum (SUVR_{Str}) also exhibited significant group differences (all $p < 0.001$), with higher uptake values observed in the A β ⁺MCI

and A β ⁺D groups compared to the A β ⁻CN group. In contrast, SUVR_{SG} displayed a significantly different pattern across groups, with median values of 2.23 (1.97–2.58) in the A β ⁻CN group, 2.04 (1.88–2.25) in the A β ⁺MCI group, and 2.09 (1.78–2.3) in the A β ⁺D group ($p=0.0049$, Fig. 1B). Vol_{SG} did not show significant group differences ($p=0.6023$).

Correlation analysis

Pearson correlation analysis demonstrated significant associations between BMI, SUVR_{SG}, and SUVR_B. BMI showed a significant negative correlation with SUVR_B ($r = -0.247$, $p = 0.0009$, Fig. 2A), indicating lower brain amyloid deposition in individuals with higher BMI. A significant positive correlation was found between BMI and SUVR_{SG} ($r = 0.231$, $p = 0.0019$, Fig. 2B). SUVR_{SG} was negatively correlated with SUVR_B ($r = -0.188$, $p = 0.0102$, Fig. 2C). Additionally, BMI exhibited a strong positive correlation with Vol_{SG} ($r = 0.469$, $p < 0.001$, Fig. 2D).

Sex-stratified Pearson correlation analyses were performed. In both males and females, BMI was significantly negatively correlated with SUVR_B (male: $r = -0.372$, $p = 0.0061$; female: $r = -0.196$, $p = 0.0287$) and positively correlated with Vol_{SG} (male: $r = 0.533$, $p < 0.001$; female: $r = 0.458$, $p < 0.001$). A significant positive correlation between BMI and SUVR_{SG} was observed in females only ($r = 0.285$, $p = 0.0013$), while the correlation was not significant in males ($r = 0.151$, $p = 0.2802$). Likewise, SUVR_{SG} was significantly negatively correlated with SUVR_B

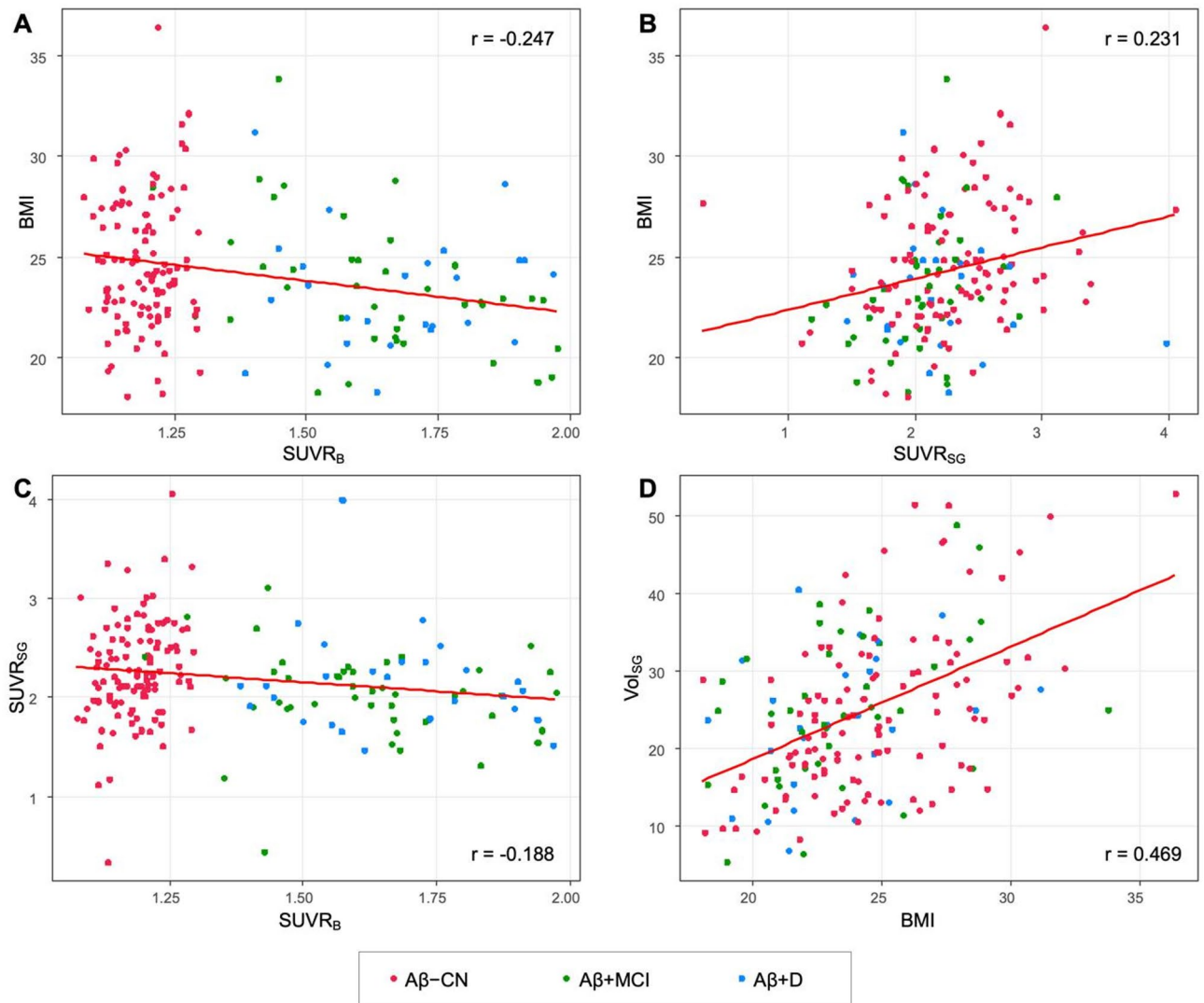


Fig. 2. Correlations between BMI, SUVR_B, SUVR_{SG}, and Vol_{SG}. (A) Body mass index (BMI) was significantly negatively correlated with SUVR_B ($r = -0.247$, $p = 0.0009$), suggesting that individuals with higher BMI tend to have lower cerebral amyloid deposition. (B) BMI showed a significant positive correlation with SUVR_{SG} ($r = 0.231$, $p = 0.0019$), indicating that increased peripheral adipose metabolism may be linked to higher amyloid uptake in the salivary glands. (C) SUVR_{SG} was negatively correlated with SUVR_B ($r = -0.188$, $p = 0.0102$), supporting the hypothesis that amyloid sequestration in peripheral tissues may be associated with lower cerebral amyloid burden. (D) BMI exhibited a strong positive correlation with salivary gland volume (Vol_{SG}) ($r = 0.469$, $p < 0.001$), consistent with previous findings that salivary gland fat content increases with BMI. Pearson correlation analysis was used for statistical comparisons.

Region	Predictor	Estimate (β)	95% CI	p-value
SUVR _B	BMI	-0.0168	-0.0283, -0.0052	0.0046*
	SUVR _{SG}	-0.0714	-0.1487, 0.0059	0.0701
	Age	0.0024	-0.0030, 0.0078	0.3838
SUVR _F	BMI	-0.0181	-0.0302, -0.006	0.0037*
	SUVR _{SG}	-0.0812	-0.1624, -0.0001	0.0498
	Age	0.0024	-0.0033, 0.0081	0.4153
SUVR _{PCu}	BMI	-0.0212	-0.0363, -0.0061	0.0061*
	SUVR _{SG}	-0.1300	-0.2311, -0.0289	0.012
	Age	0.0038	-0.0034, 0.0109	0.2989
SUVR _{Cg}	BMI	-0.0192	-0.0325, -0.0059	0.005*
	SUVR _{SG}	-0.1134	-0.2028, -0.0240	0.0132
	Age	0.0033	-0.0030, 0.0096	0.3059
SUVR _{LP}	BMI	-0.0166	-0.0288, -0.0044	0.008
	SUVR _{SG}	-0.0612	-0.1429, 0.0204	0.1407
	Age	0.0029	-0.0029, 0.0086	0.3224
SUVR _{LT}	BMI	-0.0151	-0.0256, -0.0046	0.0051*
	SUVR _{SG}	-0.0457	-0.1161, 0.0247	0.202
	Age	0.0025	-0.0024, 0.0075	0.3201
SUVR _{OC}	BMI	-0.0109	-0.0186, -0.0032	0.0058*
	SUVR _{SG}	0.0026	-0.0490, 0.0541	0.9223
	Age	0.0021	-0.0015, 0.0057	0.2511
SUVR _{Str}	BMI	-0.0105	-0.0193, -0.0017	0.0200
	SUVR _{SG}	-0.0800	-0.1391, -0.0209	0.0082
	Age	-0.0008	-0.0050, 0.0033	0.6982

Table 3. Multiple regression analysis of total and regional amyloid SUVR. BMI body mass index, SUVR_{SG} standardized uptake value ratio of salivary gland, Vol_{SG} volume of salivary gland, SUVR_B global standardized uptake value ratio of brain, SUVR_F standardized uptake value ratio of frontal cortex, SUVR_{PCu} standardized uptake value ratio of precuneus, SUVR_{Cg} standardized uptake value ratio of cingulate cortex, SUVR_{LP} standardized uptake value ratio of lateral parietal cortex, SUVR_{LT} standardized uptake value ratio of lateral temporal cortex, SUVR_{OC} standardized uptake value ratio of occipital cortex, SUVR_{Str} standardized uptake value ratio of striatum.

in females ($r = -0.249$, $p = 0.0046$), but not in males ($r = -0.066$, $p = 0.6192$). Full statistical results are summarized in Supplementary Table S1, and scatter plots for each sex are shown in Supplementary Figure S1.

Multiple regression analysis

To assess the independent contributions of BMI and SUVR_{SG} to brain amyloid burden with age as a covariate, multiple regression analyses were conducted for composite and regional brain SUVR values. The results are summarized in Table 3.

For composite brain SUVR, higher BMI was significantly associated with lower SUVR_B ($p = 0.0046$), while SUVR_{SG} showed a trend toward reduced SUVR_B, though it did not reach statistical significance ($p = 0.0701$).

For regional SUVRs, after Bonferroni correction for multiple comparisons (adjusted threshold $p < 0.0071$), BMI remained significantly associated with lower SUVR in the frontal cortex ($\beta = -0.0181$, $p = 0.0037$), precuneus ($\beta = -0.0212$, $p = 0.0061$), cingulate cortex ($\beta = -0.0192$, $p = 0.0050$), lateral temporal cortex ($\beta = -0.0151$, $p = 0.0051$), and occipital cortex ($\beta = -0.0109$, $p = 0.0058$). Associations in the lateral parietal cortex ($\beta = -0.0166$, $p = 0.0080$) and striatum ($\beta = -0.0105$, $p = 0.0200$) did not survive correction.

None of the associations between SUVR_{SG} and regional SUVR values survived Bonferroni correction. However, nominal negative associations ($p < 0.05$) were observed in the frontal cortex ($\beta = -0.0812$, $p = 0.0498$), precuneus ($\beta = -0.1300$, $p = 0.0120$), cingulate cortex ($\beta = -0.1134$, $p = 0.0132$), and striatum ($\beta = -0.0800$, $p = 0.0082$). This inverse pattern did not reach the adjusted significance threshold and should be interpreted with caution.

Discussion

Higher BMI showed a significant association with reduced cerebral amyloid burden (SUVR_B) while elevating salivary gland amyloid uptake (SUVR_{SG}), revealing an inverse relationship between these two compartments. These findings align with earlier reports demonstrating that individuals with elevated BMI exhibit lower cerebral A β deposition across PET and CSF biomarkers^{21,22}. Longitudinal evidence further corroborates this inverse relationship, with baseline A β levels predicting subsequent BMI decline²³.

A striking paradox emerges from our data: while higher BMI correlates with reduced cerebral amyloid burden, obesity and visceral adiposity simultaneously increase Alzheimer's disease risk, promote cortical

thinning²⁴ and drive neuroinflammation alongside insulin resistance²⁵. This apparent contradiction suggests that adiposity influences AD pathogenesis through distinct, potentially competing pathways.

The peripheral sink hypothesis offers a compelling mechanistic framework for these observations^{26,27}. Increased peripheral amyloid retention, as evidenced by elevated SUVR_{SG} in our cohort, may actively sequester A β away from the brain, thereby reducing cerebral burden. Supporting this model, salivary gland fat content increases proportionally with BMI, while transgenic APP mouse models demonstrate A β accumulation in salivary and lacrimal glands²⁸ establishing these peripheral tissues as functionally relevant amyloid reservoirs distinct from cerebral pathology.

Simultaneously, adipose-derived inflammatory mediators and metabolic dysfunction may accelerate neurodegeneration through amyloid-independent mechanisms²⁹. This dual pathway model reconciles the seemingly contradictory protective effect of BMI on amyloid burden with its established role in AD risk.

Region-specific analyses revealed that BMI's inverse association with SUVR_B remained robust across frontal, precuneus, cingulate, lateral temporal, and occipital cortices after Bonferroni correction. While the SUVR_B-SUVR_{SG} association reached only nominal significance ($p < 0.05$) and failed to survive multiple comparison correction, this pattern nonetheless supports the peripheral retention hypothesis.

Particularly intriguing were the nominal inverse associations between SUVR_{SG} and SUVR_B within the frontal cortex, precuneus, cingulate cortex, and striatum—regions consistently implicated in both metabolic regulation and AD pathology. The precuneus serves as a critical site for early amyloid accumulation³⁰ with visceral adiposity specifically linked to enhanced pathology in this region among middle-aged men³¹. The cingulate cortex mediates both cognitive function and metabolic control^{32,33} while the striatum bears significant impact from obesity and insulin resistance³⁴ and demonstrates amyloid pathology in advanced AD stages³⁵. Although these exploratory associations require cautious interpretation, they may reflect region-specific patterns in peripheral amyloid redistribution.

Sex-stratified analyses revealed that SUVR_{SG} associations with both BMI and SUVR_B achieved statistical significance exclusively in women. This pattern mirrors previous observations that women exhibit stronger microglial activation, a key component of neuroinflammation, independent of A β burden, particularly during early AD stages³⁶. Additionally, visceral adiposity-related biomarkers including VAT, CRP, and PAI-1 demonstrate more pronounced associations with AD pathology in women³⁷. However, the female-predominant sample composition (more than twice the male participants) may have influenced these sex-specific findings.

These results carry important implications for AD risk stratification and therapeutic development. The peripheral sink mechanism suggests that interventions targeting peripheral amyloid clearance might complement traditional brain-focused approaches. Furthermore, the sex-specific patterns indicate that metabolic interventions for AD prevention may require gender-tailored strategies.

The novel application of SUVR_{SG} as a PET-based peripheral amyloid marker represents a key methodological advance. To our knowledge, this constitutes the first in vivo investigation examining salivary gland amyloid uptake in relation to cerebral A β burden. Integrating central and peripheral amyloid PET measures enabled a more comprehensive assessment of systemic A β dynamics, while region-specific SUVR_B analyses provided spatially nuanced insights into adiposity's influence on distinct cortical regions.

Nevertheless, several limitations warrant acknowledgment. The cross-sectional design precludes causal inference, necessitating longitudinal validation to establish temporal relationships. Although SUVR_{SG} served as a proxy for peripheral amyloid retention, its specificity for adipose-related A β accumulation awaits histopathological confirmation. The female-predominant sample may have skewed interpretations beyond sex-stratified analyses. Finally, incomplete control for potential confounders including comorbidities, medications, and lifestyle factors may have introduced residual bias, which is inherent to our retrospective study design.

This investigation underscores the critical role of peripheral compartments in amyloid distribution and highlights metabolic factors' complex influence on AD pathology. By incorporating SUVR_{SG} into our analytical framework, we provide mechanistic insights into how adiposity may modulate brain A β burden through peripheral sequestration. Key questions remain for advancing our understanding of adiposity's role in dementia. It is unclear whether this peripheral redistribution mechanism applies to other A β -associated conditions such as cerebral amyloid angiopathy or Down syndrome-related dementia, but doing so could uncover generalizable protective mechanisms across the A β spectrum. Integrating adiposity indices with accessible fluid biomarkers, including emerging plasma A β 42/40 ratios and salivary amyloid measures, may help validate PET findings in more routine clinical contexts. Longitudinal studies that monitor peripheral and cerebral amyloid dynamics in parallel will be essential to confirm the peripheral sink hypothesis and explore its therapeutic potential.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

K.R.K, K.M.K: Conception and design of study, acquisition of data, data curation, formal analysis. H.J.Y, J.Y.K: Conception and design of study, data curation, formal analysis. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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